

Tiernen Bunnell

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EDUCATION

Western Governors University — B.S. in Computer Science

Expected February 2027

TECHNICAL SKILLS

Languages: Go, Python, Java, SQL, JavaScript

Backend & APIs: REST APIs, HTTP, Spring Boot, Spring Data JPA, HTML, CSS

Cloud & Infrastructure: AWS, Docker, Docker Compose, Terraform, Linux, Git, GitHub Actions, CI/CD

Databases & Observability: PostgreSQL, Redis, MySQL, MariaDB, Prometheus, Grafana, CloudWatch

Core: Data Structures & Algorithms, Distributed Systems, Concurrency, Object-Oriented Design, Networking, Software Testing

PROJECTS

Data Processing Platform

Go · Python · PostgreSQL · Redis · Docker · AWS · Terraform · [Live Demo](#)

- Built and deployed a full-stack asynchronous data-processing platform in Go and Python for dataset analysis, image processing, and route optimization using PostgreSQL and Redis Streams.
- Designed concurrent worker infrastructure with Redis consumer groups, atomic PostgreSQL job claiming, pending-job recovery, bounded retries, timeouts, and queue backpressure; benchmarked near-linear throughput scaling (9.9 → 29.6 jobs/sec across 1–3 workers).
- Provisioned and deployed AWS infrastructure with Terraform across EC2, RDS, S3, ECR, IAM, Secrets Manager, and CloudWatch; automated ARM64 container publishing and EC2 deployment with GitHub Actions and AWS Systems Manager.
- Implemented authentication and authorization with guest sessions and per-user job isolation; added Prometheus/Grafana metrics, CloudWatch logging, and live JavaScript job-status updates.

Homelab Dashboard

Go · Docker · REST APIs · JavaScript

- Built and published a full-stack Go web application exposing REST APIs to monitor and control Docker/Podman containers through the Docker Engine API.
- Designed a background polling goroutine with an in-memory, RWMutex-protected cache to serve live container and host resource metrics without repeatedly querying the Docker API.
- Built GitHub Actions CI/CD to run tests and vetting, build and publish Docker images to GHCR, and deploy through a self-hosted runner with post-deployment health checks.

WGUPS Routing Optimization

Python · Algorithms · Data Structures

- Built a delivery simulation for 40 packages with deadlines, capacity limits, delayed arrivals, address corrections, and grouping constraints.
- Implemented a custom hash table with average $O(1)$ insertion, lookup, and update operations.
- Developed a routing algorithm using Nearest Neighbor and 2-opt heuristics to schedule deliveries under multiple constraints.

CO₂ Prediction & Model Optimization

Python · scikit-learn

- Built regression models to predict CO₂ levels from an air-quality dataset after cleaning and feature selection.
- Compared decision tree, random forest, and gradient boosting models using MSE, RMSE, and R² with randomized and grid hyperparameter searches.
- Analyzed model performance and feature relationships through visualizations and documented methodology, results, limitations, and improvements.

EXPERIENCE

Cracker Barrel — Dishwasher / Cook

August 2020 – February 2022

- Worked reliably in a high-volume environment while balancing multiple responsibilities, coordinating with kitchen staff, maintaining accuracy under time constraints, and communicating effectively with teammates.